

Decision and Risk Analysis

Lecture 4: **Bayesian Balance and Sequential Learning**

DAEN 427 / ISEN 427 / ISEN 627

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Learning objectives

By the end of class, you should be able to:

- ▶ explain how prior information and observed data jointly determine a posterior;
- ▶ quantify their influence in a Beta–Binomial model;
- ▶ update beliefs sequentially as evidence arrives;
- ▶ recognize when a prior prevents learning from evidence;
- ▶ connect posterior probabilities to risk decisions.

Decision setting

Let π denote the unknown probability of an event.

- ▶ Safety: probability that a component fails during a mission.
- ▶ Operations: probability that demand exceeds capacity.
- ▶ Screening: probability that a sampled item is nonconforming.

We observe Y events in n comparable trials:

$$Y \mid \pi \sim \text{Bin}(n, \pi), \quad \pi \sim \text{Beta}(\alpha, \beta).$$

The posterior describes uncertainty about the event rate. A decision still requires consequences, alternatives, and a decision rule.

Running example: Bechdel test

The chapter asks what proportion π of recent films pass a representation test. Three analysts begin with different beliefs.

Analyst	Prior	Interpretation
Skeptic	Beta(5, 11)	pass rates below 0.5 seem more plausible
Uncertain	Beta(1, 1)	all rates receive equal density
Optimist	Beta(14, 1)	rates near 1 seem highly plausible

Which prior carries the most information?

How can you tell without seeing a graph?

Reading Beta priors

For $\pi \sim \text{Beta}(\alpha, \beta)$:

$$E(\pi) = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(\pi) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

- ▶ The ratio $\alpha : (\alpha + \beta)$ locates the prior mean.
- ▶ The total $\alpha + \beta$ measures prior concentration and behaves like a prior sample size.
- ▶ A vague prior spreads plausibility broadly; an informative prior concentrates it.

Prior concentration determines how much new evidence a model needs before a risk estimate moves substantially.

R: inspect competing priors

```
library(tidyverse)

priors <- tibble(
  analyst = c("Skeptic", "Uncertain", "Optimist"),
  alpha = c(5, 1, 14),
  beta = c(11, 1, 1)
)

grid <- crossing(priors, pi = seq(0, 1, length.out = 501)) |>
  mutate(density = dbeta(pi, alpha, beta))

ggplot(grid, aes(pi, density, color = analyst)) +
  geom_line(linewidth = 1) + theme_minimal()
```

Interpretation (before computation!):

location represents belief; spread represents certainty.

Conjugate updating keeps the calculation simple

After observing $Y = y$ successes in n trials,

$$\pi \mid (Y = y) \sim \text{Beta}(\alpha + y, \beta + n - y).$$

The update has a useful counting interpretation:

posterior success parameter = α + observed successes,

posterior failure parameter = β + observed failures.

Note: α and β are hyperparameters.

Calling them literal past observations is often only an analogy.

Same data, different posteriors

In a sample of 20 films, 9 pass. Every analyst sees the same likelihood.

Analyst	Prior	Posterior	Posterior mean
Skeptic	Beta(5, 11)	Beta(14, 22)	0.389
Uncertain	Beta(1, 1)	Beta(10, 12)	0.455
Optimist	Beta(14, 1)	Beta(23, 12)	0.657

- ▶ The vague prior yields a posterior centered close to $9/20 = 0.45$.
- ▶ The concentrated optimistic prior still pulls the posterior upward.

With limited evidence, plausible prior judgments can produce materially different risk estimates. Sensitivity analysis should show that range.

Same prior, similar rate, different sample sizes

All three analysts use $\text{Beta}(14, 1)$ and observe a pass rate near 46%.

Data	Sample rate	Posterior	Posterior mean
6/13	0.462	$\text{Beta}(20, 8)$	0.714
29/63	0.460	$\text{Beta}(43, 35)$	0.551
46/99	0.465	$\text{Beta}(60, 54)$	0.526

The larger sample generates a narrower likelihood and pulls the posterior farther from the optimistic prior.

Evidence quality involves more than an observed rate. Sample size controls how strongly the rate should change a decision model.

Posterior mean as a weighted compromise

$$E(\pi \mid y) = \frac{\alpha + y}{\alpha + \beta + n} = \underbrace{\frac{\alpha + \beta}{\alpha + \beta + n}}_{w_{\text{prior}}} \underbrace{\frac{\alpha}{\alpha + \beta}}_{\text{prior mean}} + \underbrace{\frac{n}{\alpha + \beta + n}}_{w_{\text{data}}} \underbrace{\frac{y}{n}}_{\text{sample rate}}.$$

$$w_{\text{prior}} + w_{\text{data}} = 1.$$

This identity makes the balance explicit. The prior contributes $\alpha + \beta$ units of concentration; the data contribute n trials.

Worked balance calculation

Use the optimistic prior $\text{Beta}(14, 1)$.

Small sample: 6/13

$$w_{\text{prior}} = \frac{15}{28} = 0.536$$

$$\begin{aligned} E(\pi | y) &= 0.536(0.933) + 0.464(0.462) \\ &= 0.714. \end{aligned}$$

Larger sample: 46/99

$$w_{\text{prior}} = \frac{15}{114} = 0.132$$

$$\begin{aligned} E(\pi | y) &= 0.132(0.933) + 0.868(0.465) \\ &= 0.526. \end{aligned}$$

The same prior and almost the same observed rate lead to different posterior means because the evidence has different weight.

What happens as evidence accumulates?

For a fixed proper Beta prior,

$$\frac{\alpha + \beta}{\alpha + \beta + n} \longrightarrow 0, \quad \frac{n}{\alpha + \beta + n} \longrightarrow 1.$$

Therefore,

$$E(\pi \mid y) \longrightarrow \frac{y}{n} \quad \text{as } n \text{ grows.}$$

Analysts with different non-stubborn priors tend to converge when they observe strong common evidence.

Does convergence guarantee a good decision if the trials are biased or not comparable to the decision context?

- ▶ More data do not automatically mean better inference.
- ▶ If the observations are biased, dependent, measured incorrectly, or drawn from a population that differs from the decision population, the posterior may become extremely certain about the wrong value.
- ▶ Bayesian updating correctly processes the evidence described by the model; it cannot by itself correct a flawed sampling or data-generating process.

R: visualize and summarize the balance

```
library(bayesrules)

plot_beta_binomial(alpha = 14, beta = 1,
                   y = 6, n = 13)

summarize_beta_binomial(alpha = 14, beta = 1,
                       y = 6, n = 13)

# Compare with stronger evidence at nearly the same rate
plot_beta_binomial(alpha = 14, beta = 1,
                   y = 46, n = 99)
```

Look for three objects: prior density, scaled likelihood, and posterior density. A scaled likelihood aids visual comparison but remains distinct from a probability density.

From probability to a risk decision

Suppose action a_1 mitigates an adverse event and action a_0 does not. Let $L(a, \pi)$ denote loss.

$$\text{Posterior expected loss}(a) = E[L(a, \pi) \mid y].$$

Choose the action with smaller posterior expected loss.

Simple threshold case

If mitigation costs C and an unmitigated event causes loss D , mitigate when

$$P(\text{event} \mid y) D > C.$$

Prior sensitivity matters most when competing posterior decisions lie near the action threshold.

Sequential Bayesian learning

As each batch arrives, yesterday's posterior becomes today's prior.

$$\text{Beta}(\alpha_{t-1}, \beta_{t-1}) \xrightarrow{y_t \text{ of } n_t} \text{Beta}(\alpha_{t-1} + y_t, \beta_{t-1} + n_t - y_t).$$

This supports risk monitoring in settings where evidence arrives over time: reliability tests, incident counts, inspections, and demand observations.

Milgram example: three sequential batches

The chapter begins with $\pi \sim \text{Beta}(1, 10)$ and updates three batches.

Stage	New data	Current model	Posterior mean
0	none	$\text{Beta}(1, 10)$	0.091
1	1/10	$\text{Beta}(2, 19)$	0.095
2	17/20	$\text{Beta}(19, 22)$	0.463
3	8/10	$\text{Beta}(27, 24)$	0.529

The second batch causes the largest change because it conflicts sharply with the belief accumulated through stage 1.

R: implement sequential updating

```
batches <- tibble(day = 1:3,  
                  y = c(1, 17, 8),  
                  n = c(10, 20, 10))  
  
alpha <- 1; beta <- 10  
history <- tibble(day = 0, alpha = alpha, beta = beta)  
  
for (i in seq_len(nrow(batches))) {  
  alpha <- alpha + batches$y[i]  
  beta <- beta + batches$n[i] - batches$y[i]  
  history <- rbind(history,  
                   tibble(day = i, alpha = alpha, beta = beta))  
}  
  
history |> mutate(mean = alpha / (alpha + beta))
```

This pattern generalizes to any stream of Binomial batches.

Order changes the path, not the destination

Original order:

$$\text{Beta}(1, 10) \rightarrow \text{Beta}(2, 19) \rightarrow \text{Beta}(19, 22) \rightarrow \boxed{\text{Beta}(27, 24)}.$$

Reverse order:

$$\text{Beta}(1, 10) \rightarrow \text{Beta}(9, 12) \rightarrow \text{Beta}(26, 15) \rightarrow \boxed{\text{Beta}(27, 24)}.$$

The intermediate beliefs differ. The final posterior depends only on cumulative totals: 26 successes and 14 failures.

Operational actions taken between batches can differ even when the final statistical model is order invariant. Timing still matters for decisions.

Why order invariance holds

For conditionally independent observations y_1, y_2 ,

$$f(\theta \mid y_1, y_2) \propto f(\theta)L(\theta \mid y_1)L(\theta \mid y_2).$$

Multiplication commutes:

$$L(\theta \mid y_1)L(\theta \mid y_2) = L(\theta \mid y_2)L(\theta \mid y_1).$$

Sequential and pooled updates therefore yield the same final posterior under the same model assumptions. **Boundary of the result:** dependence, changing data-generating processes, or adaptive sampling may require an expanded model.

A prior can block learning

Suppose a researcher assigns

$$\pi \sim \text{Unif}(0, 0.25).$$

Even after observing 8 events in 10 trials, Bayes' rule gives

$$f(\pi \mid y = 8) \propto \pi^8 (1 - \pi)^2 \quad \text{only for } 0 \leq \pi \leq 0.25.$$

The posterior cannot place probability above 0.25 because the prior assigns that region probability zero.

Modeling rule

Avoid excluding technically possible parameter values unless the exclusion follows from a genuine physical or logical constraint.

Subjectivity and transparent analysis

Bayesian analysis makes several judgments visible:

- ▶ prior location and concentration;
- ▶ likelihood and sampling assumptions;
- ▶ outcomes, losses, and decision thresholds.

Good practice for risk analysis:

1. justify the prior from data, expertise, or a stated weak-information goal;
2. report sensitivity to plausible alternatives;
3. inspect prior predictive implications;
4. revise the model when the evidence reveals misspecification.

In-class

A maintenance action costs \$12,000. A failure costs \$100,000. Assume the action prevents failure and has no other consequence.

1. What posterior failure probability triggers maintenance?
2. Would the decision change under posterior means 0.09, 0.14, and 0.22?
3. What important consequences or uncertainties does this simple rule omit?

In-class (1)

A maintenance action costs \$12,000. Failure costs \$100,000. Assume maintenance completely prevents failure and has no other consequence.

Let $p = P(\text{failure} \mid y)$. Compare posterior expected costs:

$$E[C \mid \text{no maintenance}, y] = 100,000p,$$

$$E[C \mid \text{maintenance}, y] = 12,000.$$

Maintenance has lower expected cost when

$$100,000p > 12,000 \iff \boxed{p > 0.12}.$$

At $p = 0.12$, the two actions have equal expected cost.

In-class: (2) and (3)

Posterior failure probability	Expected cost without maintenance	Decision
0.09	\$9,000	Do not maintain
0.14	\$14,000	Maintain
0.22	\$22,000	Maintain

What the simple model omits

- ▶ Maintenance may reduce rather than eliminate failure risk.
- ▶ Failure losses and maintenance effectiveness may be uncertain.
- ▶ Downtime, safety, environmental, and reputational consequences may matter.
- ▶ Risk aversion and the value of waiting for more information may change the choice.

Conclusions

- ▶ The posterior balances the prior and evidence.
- ▶ Prior concentration and sample size determine the balance in the Beta–Binomial model.
- ▶ Sequential updating reuses the current posterior as the next prior.
- ▶ The final posterior is order invariant under the stated model assumptions.
- ▶ Zero prior support cannot be repaired by contradictory data.
- ▶ Risk decisions require posterior uncertainty plus explicit consequences.

Homework

1. **Exercise 4.3** Propose reasonable Beta priors for five judgments about the chance that a ginkgo tree drops its leaves next Monday: very unlikely; completely unsure; very likely based on last year; a decent chance with uncertainty; probably not, with uncertainty. Justify each choice using its mean and concentration.
2. **Exercise 4.7** For each of the five prior-and-data scenarios in the chapter, determine whether the prior, data, or neither has greater influence. Verify using w_{prior} and w_{data} and plot at least two cases in R.

Homework

3. **Exercise 4.13** Analyze a politician's $\text{Unif}(0.5, 1)$ prior after a poll reports 0 approvals among 100 people. Sketch the prior and posterior, describe the conclusion, and diagnose the modeling error.
4. **Exercise 4.18** Begin with $\text{Beta}(4, 3)$ for an ad click rate. Update sequentially with batches $(y, n) = (0, 1), (3, 10), (20, 100)$. Report and plot each posterior, then verify that pooling all data gives the same final posterior.

Include in the submission: derivations, interpretations in complete sentences, and reproducible R code (with labeled plots).